scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

Introduction

Single-cell RNA sequencing jointly captures spliced and unspliced transcripts, yet computational frameworks have focused almost exclusively on transcriptional dynamics—most notably RNA velocity, which models the rate of change of spliced mRNA abundance [1,2]. Meanwhile, mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs, and *cis*-regulatory elements—is a major determinant of steady-state gene expression and is increasingly recognized as a driver of cell fate transitions [3]. Despite this, no general framework exists to systematically estimate per-cell mRNA degradation rates, discover cell populations defined by post-transcriptional programs, or infer the regulatory networks governing mRNA stability at single-cell resolution. We present **scPTR** (Single-Cell Post-Transcriptional Regulatory Decomposition), a framework that estimates per-cell, per-gene mRNA degradation rates (γ) from standard scRNA-seq data and treats γ as a primary analytical axis—complementary to, and independent of, RNA velocity.

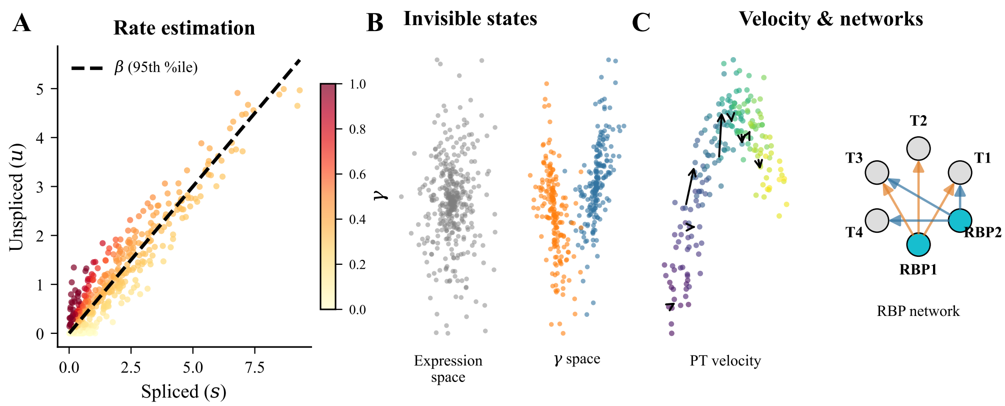

Figure 1: scPTR framework. **(A)** Gene-specific splicing rates (β) are estimated via quantile regression on unspliced/spliced phase portraits; per-cell degradation rates are then computed as $\gamma = \beta \cdot u/s$ after k-NN Gaussian-kernel smoothing. **(B)** Clustering in γ-space via PCA and Leiden community detection reveals post-transcriptional states that are invisible to standard expression-based analysis. **(C)** Post-transcriptional velocity (the neighbor-averaged γ gradient) captures degradation dynamics orthogonal to RNA velocity, while library-size-corrected elastic net regression infers RBP–target regulatory networks.

Methods

For each gene, scPTR estimates the splicing rate β as the 95th-percentile slope of the unspliced/spliced phase portrait (Fig. 1A), then computes per-cell degradation rates via the kinetic steady-state relation $\gamma_{ig} = \beta_g \cdot u_{ig}/s_{ig}$, where u and s are k-NN Gaussian-kernel-smoothed unspliced and spliced counts, respectively. The smoothing step uses adaptive bandwidths (median neighbor distance per cell) to denoise inherently sparse unspliced counts without oversmoothing. Post-transcriptional (PT) states are discovered by performing PCA, k-NN graph construction, and Leiden clustering entirely in γ-space rather than expression space (Fig. 1B). PT velocity is defined as the weighted mean γ difference between each cell and its neighbors, yielding a per-cell vector field orthogonal to RNA velocity (Fig. 1C). RBP–target regulatory networks are inferred via elastic net regression of each target gene's γ on smoothed RBP expression, with library-size partial correlation to correct for a systematic destabilizing bias arising from the correlation between library size and count-based γ estimates. We additionally introduce DeepPTR, a structured variational autoencoder with a kinetic decoder that factorizes the latent space into transcriptional (z_T) and post-transcriptional (z_{PT}) components under a negative binomial observation model.

Results

We validated γ estimates against published mRNA half-lives across three species–platform combinations, achieving Spearman correlations from -0.33 (10x mouse developmental data) to -0.81 (sci-fate human metabolic labeling with 4sU; Fig. 2A). The strongest correlations arise in sci-fate because metabolic labeling captures new/old RNA directly, providing a cleaner unspliced signal, and permits same-species comparison with the human half-life reference. Predicted miRNA targets from TargetScan 8.0 were enriched among high-γ genes for 59% of 215 tested families (Fig. 2B), and per-gene γ correlated positively with both $3'$ UTR length ($\rho = 0.34$, $p < 10^{-200}$) and AU content ($\rho = 0.30$, $p < 10^{-150}$), consistent with known cis-regulatory determinants of mRNA stability. Gamma estimates were also robust to cell subsampling ($r > 0.97$ at 20% subsampling) and exhibited sub-linear runtime scaling ($\sim$91 seconds estimated for 100,000 cells).

Clustering in γ-space revealed expression-invisible PT states in 3 of 8 pancreatic and 6 of 11 hippocampal cell types—subpopulations with distinct degradation programs yielding higher silhouette scores in γ-space than expression space (Fig. 3A). In pancreas, the Epsilon cluster showed the strongest invisibility: its gamma sub-clusters had silhouette $= 0.195$ in γ-space but were actively anti-clustered in expression space (silhouette $= -0.056$). These invisible states were enriched for biologically coherent pathways via

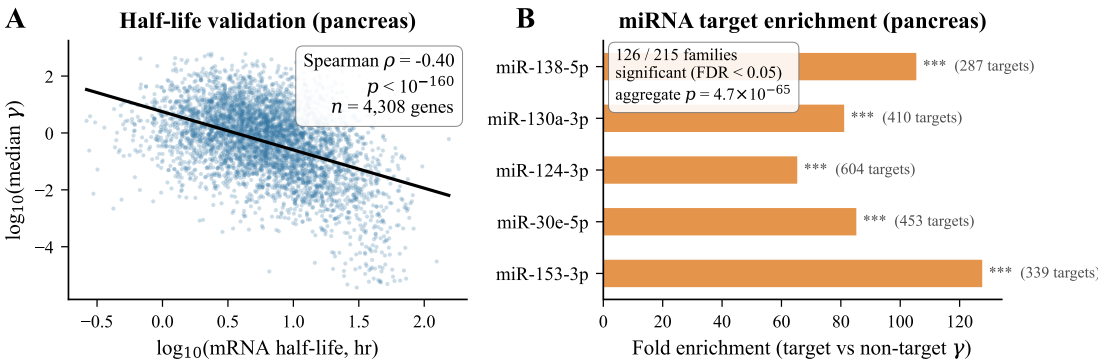

Figure 2: Validation of γ estimates. **(A)** Per-gene median γ correlates negatively with published mRNA half-lives in the pancreas dataset (Spearman $\rho = -0.40$, $n = 4{,}308$ genes; $\rho = -0.81$ in sci-fate metabolic-labeling data). **(B)** Top miRNA families show strong fold enrichment of γ among predicted targets vs. non-targets in pancreas (126/215 families significant at FDR < 0.05; aggregate $p = 4.7 \times 10^{-65}$).

GSEA: ER protein processing and autophagy in pancreatic endocrine cells; synaptic plasticity and spliceosome in hippocampal neurons—pathways directly relevant to the known biology of secretory and neuronal cells.

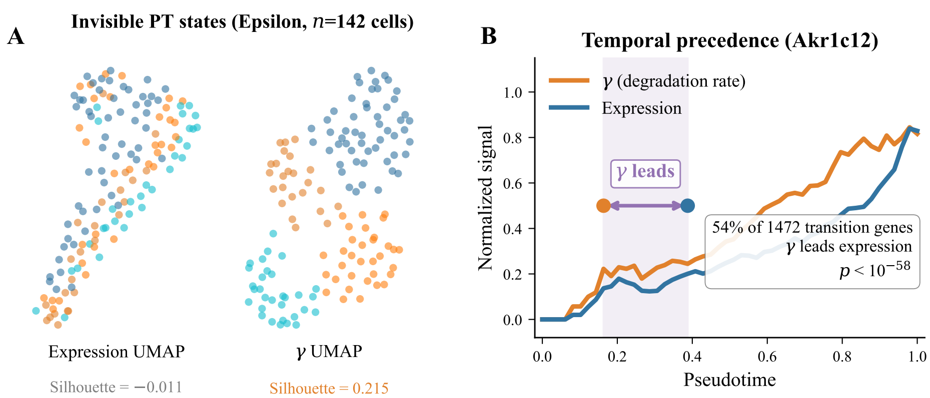

Figure 3: Key biological findings. **(A)** Pancreas Epsilon cells ($n = 142$) appear as one homogeneous cluster in expression-space UMAP (silhouette $= -0.011$) but separate into distinct post-transcriptional states in γ-space UMAP (silhouette $= 0.215$), demonstrating expression-invisible heterogeneity in mRNA degradation programs. **(B)** Along developmental pseudotime, changes in γ (red) precede changes in expression (blue) for 54% of 1,472 transition genes ($p < 10^{-57}$), shown here for *Akr1c12*. Post-transcriptional rewiring acts as an early fate-commitment signal.

PT velocity—the neighbor-averaged γ gradient—was orthogonal to RNA velocity (mean cosine similarity ≈ 0) and demonstrated that degradation-rate changes temporally precede expression changes for 63–78% of transition genes during differentiation (Fig. 3B), with a mean lag of 2–4 pseudotime bins. This temporal ordering was significant in both pancreas ($p = 5.3 \times 10^{-6}$) and dentate gyrus ($p = 9.9 \times 10^{-13}$), supporting the hypothesis that post-transcriptional regulation acts as an early signal during cell fate transitions. Library-size-corrected RBP–target network inference identified hub regulators including HNRNPA1, YBX1, and ELAVL1/HuR, whose functional importance was validated by DepMap CRISPR essentiality scores (hub RBPs significantly more essential; $p = 6.4 \times 10^{-5}$). Application to human neuroblastoma [4] revealed a predominantly stabilizing RBP landscape (66% stabilizing edges after correction), contrasting sharply with the destabilizing bias observed in developmental tissues and consistent with oncogenic mRNA stabilization programs. Hub RBPs in neuroblastoma (HNRNPA2B1, PABPC1) matched known cancer dependencies [5].

Conclusions

scPTR reveals a systematic, previously inaccessible layer of post-transcriptional regulation in single-cell data. By treating mRNA degradation rates as per-cell observables, scPTR discovers expression-invisible cell states, demonstrates that post-transcriptional rewiring precedes transcriptional changes during fate transitions, and identifies functionally essential RBP regulators—establishing degradation-rate analysis as a complement to RNA velocity for single-cell biology.

Availability: `https://github.com/bryanc5864/scPTR`

References

[1] La Manno et al. RNA velocity of single cells. *Nature* **560**, 494–498 (2018). [2] Bergen et al. Generalizing RNA velocity to transient cell states. *Nat. Biotechnol.* **38**, 1408–1414 (2020).

[3] Hentze et al. A brave new world of RNA-binding proteins. *Nat. Rev. Mol. Cell Biol.* **19**, 327–341 (2018).

[4] Dong et al. *Cancer Cell* **38**, 716–733 (2020). [5] Dempster et al. *Genome Biol.* **22**, 343 (2021).